P2 Chapter 8: Parametric Equations

Points of Intersection

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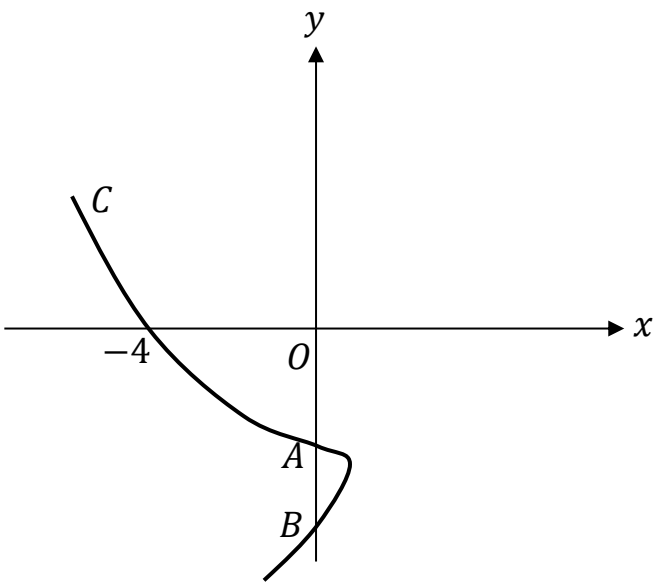
We can find where a parametric curve crosses a particular axis or where curves cross each other.

The key is to first find the value of the parameter t.

[Textbook] The diagram shows a curve C with parametric equations $x = at^2 + t, \quad y = a(t^3 + 8)$, $t \in \mathbb{R}$, where a is a non-zero constant. Given that C passes through the point $(-4,0)$,

a) find the value of a.

b) find the coordinates of the points A and B where the curve crosses the y-axis.



a

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b

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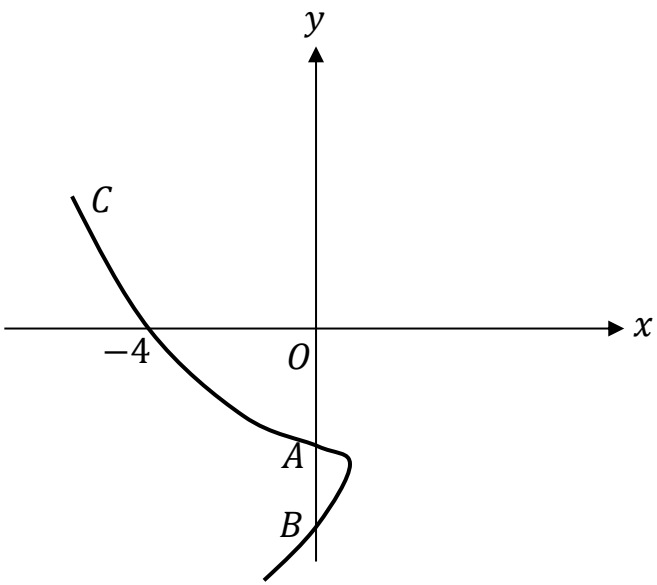

a

$$-4 = at^2 + t$$
$$0 = a(t^3 + 8)$$

From equation (2), because $a \neq 0$,

$t^3 + 8 = 0 \quad \rightarrow \quad t = -2$

Substituting into (1):

$$-4 = 4a - 2 \quad \rightarrow \quad a = -\frac{1}{2}$$

b

On the y-axis, $x = 0$:

$$0 = at^2 + t = -\frac{1}{2}t^2 + t$$

$$t\left(-\frac{1}{2}t + 1\right) = 0$$

$t = 0 \; or \; t = 2$

At $t = 0, \; y = -\frac{1}{2}(0^3 + 8) = -4 \quad \rightarrow (0, -4)$

At $t = 2, y = -\frac{1}{2}(2^3 + 8) = -8 \quad \rightarrow (0, -8)$

Points of Intersection

[Textbook] A curve is given parametrically by the equations $x = t^2,\ y = 4t$. The line $x + y + 4 = 0$ meets the curve at A. Find the coordinates of A.

$$x + y + 4 = 0$$
$$\therefore t^2 + 4t + 4 = 0$$
$$(t+2)^2 = 0 \quad \therefore \quad t = -2$$
$$\therefore x = (-2)^2 = 4$$
$$y = 4(-2) = -8$$
$$\mathrm{A}(4, -8)$$

Whenever you want to solve a Cartesian equation and pair of parametric equations simultaneously, substitute the parametric equations into the Cartesian one.

[Textbook] The diagram shows a curve C with parametric equations

$$x = \cos t + \sin t, \qquad y = \left(t - \frac{\pi}{6}\right)^2, \qquad -\frac{\pi}{2} < t < \frac{4\pi}{3}$$

a) Find the point where the curve intersects the line $y = \pi^2$.

b) Find the coordinates of the points A and B where the curve cuts the y-axis.

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a

$$\pi^2 = \left(t - \frac{\pi}{6}\right)^2 \quad \rightarrow \quad t = \frac{7}{6}\pi \ \ or \ \ -\frac{5}{6}\pi$$

(But reject $-\frac{5}{6}\pi$ as outside domain of t)

If $t = \frac{7}{6}\pi, \quad x = -\frac{1+\sqrt{3}}{2}$

$$\left(-\frac{1+\sqrt{3}}{2}, \pi^2\right)$$

b

When $x = 0$,

$\cos t + \sin t = 0$

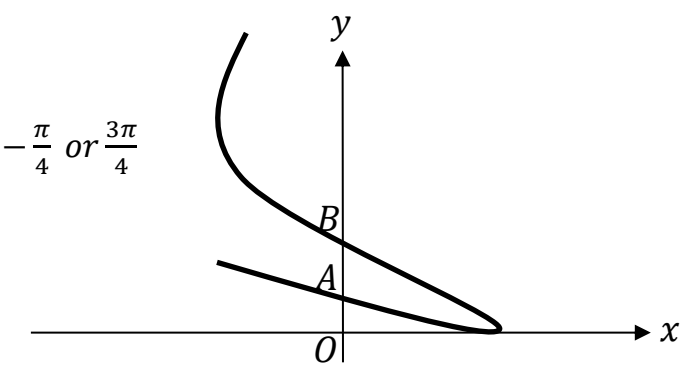
$\sin t = -\cos t$

$\tan t = -1$

Since $-\frac{\pi}{2} < t < \frac{3\pi}{4}$, $t = -\frac{\pi}{4}\ or \frac{3\pi}{4}$

$$t = -\frac{\pi}{4} \rightarrow \left(0, \frac{25\pi}{144}\right)$$

$$t = \frac{3\pi}{4} \rightarrow \left(0, \frac{121\pi}{144}\right)$$



Test Your Understanding

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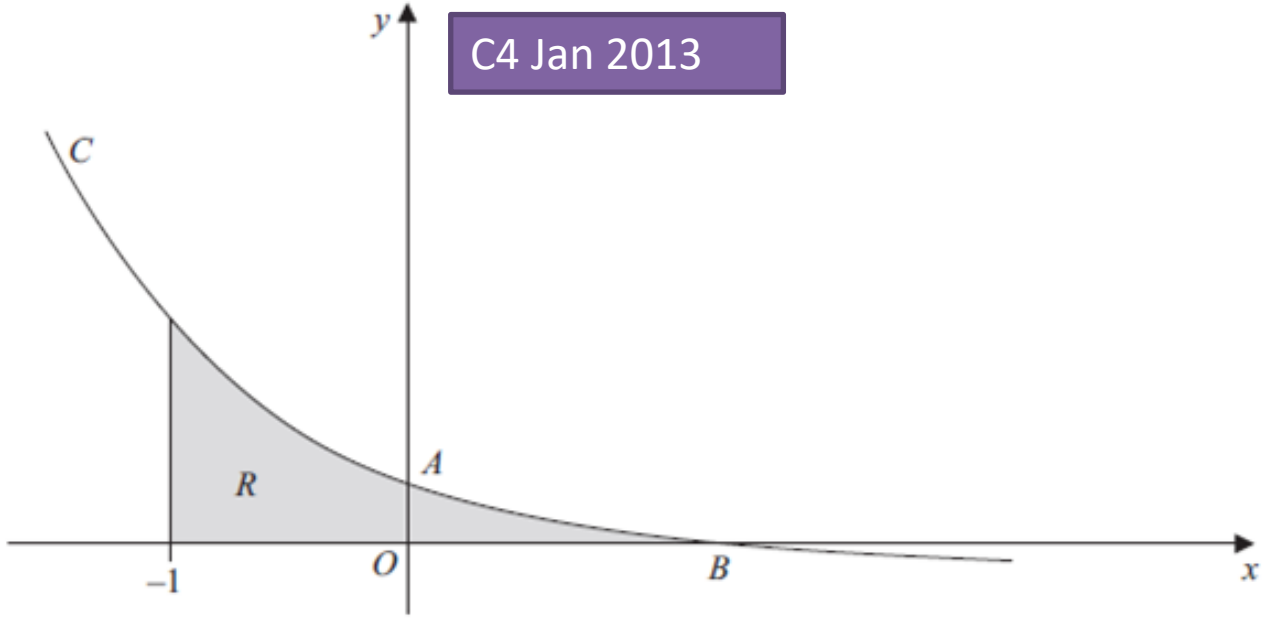

Figure 2

Figure 2 shows a sketch of part of the curve C with parametric equations

$$x = 1 - \frac{1}{2}t, \qquad y = 2^t - 1.$$

The curve crosses the y-axis at the point A and crosses the x-axis at the point B.

(*a*) Show that A has coordinates (0, 3).

(2)

(*b*) Find the x-coordinate of the point B.

(2)

a ?

b ?

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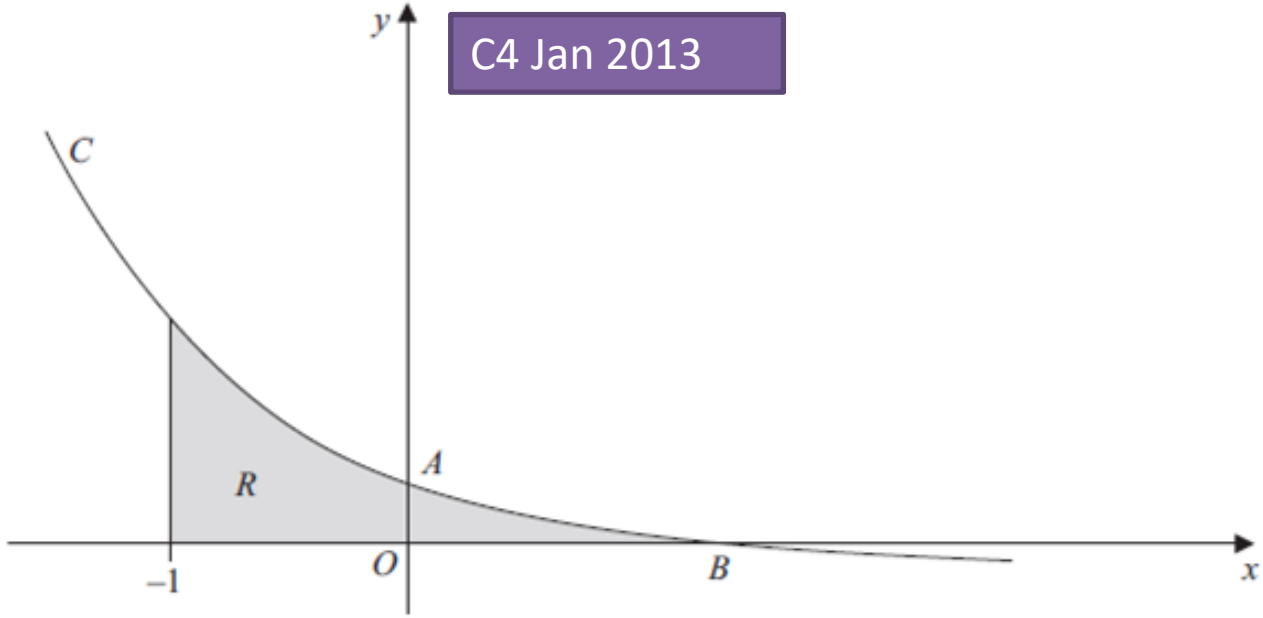

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The curve crosses the y-axis at the point A and crosses the x-axis at the point B.

(*a*) Show that A has coordinates (0, 3). **(2)**

(*b*) Find the x-coordinate of the point B. **(2)**

If $x = 0$,

$$0 = 1 - \frac{1}{2}t$$

$$t = 2$$

Then:

$$y = 2^2 - 1 = 3$$

If $y = 0$,

$$2^t - 1 = 0$$

$$t = 0$$

$$x = 1 - \frac{1}{2}(0) = 1$$

Exercise 8.4

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Homework Exercise

1 Find the coordinates of the point(s) where the following curves meet the x-axis.

a $x = 5 + t, \quad y = 6 - t$

b $x = 2t + 1, \quad y = 2t - 6$

c $x = t^2, \quad y = (1 - t)(t + 3)$

d $x = \frac{1}{t}, \quad y = (t - 1)(2t - 1), \quad t \neq 0$

e $x = \frac{2t}{1 + t}, \quad y = t - 9, \quad t \neq -1$

2 Find the coordinates of the point(s) where the following curves meet the y-axis.

a $x = 2t, \quad y = t^2 - 5$

b $x = 3t - 4, \quad y = \frac{1}{t^2}, \quad t \neq 0$

c $x = t^2 + 2t - 3, \quad y = t(t - 1)$

d $x = 27 - t^3, \quad y = \frac{1}{t - 1}, \quad t \neq 1$

e $x = \frac{t - 1}{t + 1}, \quad y = \frac{2t}{t^2 + 1}, \quad t \neq -1$

3 A curve has parametric equations $x = 4at^2, \quad y = a(2t - 1)$, where a is a constant. The curve passes through the point (4, 0). Find the value of a.

4 A curve has parametric equations $x = b(2t - 3), \quad y = b(1 - t^2)$, where b is a constant. The curve passes through the point (0, −5). Find the value of b.

5 Find the coordinates of the point of intersection of the line with parametric equations $x = 3t + 2$, $y = 1 - t$ and the line $y + x = 2$.

6 Find the values of t at the points of intersection of the line $4x - 2y - 15 = 0$ with the parabola $x = t^2, \quad y = 2t$ and give the coordinates of these points.

7 Find the points of intersection of the parabola $x = t^2$, $y = 2t$ with the circle $x^2 + y^2 - 9x + 4 = 0$.

Homework Exercise

8 Find the coordinates of the point(s) where the following curves meet the x-axis and the y-axis.

a $x = t^2 - 1, \quad y = \cos t, \quad 0 < t < \pi$

b $x = \sin 2t, \quad y = 2\cos t + 1, \quad \pi < t < 2\pi$

c $x = \tan t, \quad y = \sin t - \cos t, \quad 0 < t < \frac{\pi}{2}$

9 Find the coordinates of the point(s) where the following curves meet the x-axis and the y-axis.

a $x = e^t + 5, \quad y = \ln t, \quad t > 0$

b $x = \ln t, \quad y = t^2 - 64, \quad t > 0$

c $x = e^{2t} + 1, \quad y = 2e^t - 1, \quad -1 < t < 1$

10 Find the values of t at the points of intersection of the line $y = -3x + 2$ and the curve with parametric equations $x = t^2$, $y = t$, and give the coordinates of these points.

11 Find the value(s) of t at the point of intersection of the line $y = x - \ln 3$ and the curve with parametric equations $x = \ln(t - 1)$, $y = \ln(2t - 5)$, $t > \frac{5}{2}$, and give the exact coordinates of this point.

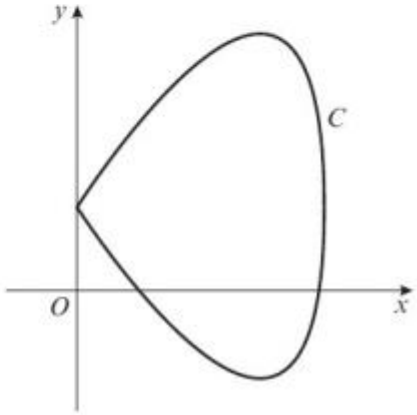
12 A curve C has parametric equations

$$x = 6\cos t, \quad y = 4\sin 2t + 2, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$$

a Find the coordinates of the points where the curve intersects the x-axis. **(4 marks)**

b Show that the curve crosses the line $y = 4$ where $t = \frac{\pi}{12}$ and $t = \frac{5\pi}{12}$ **(3 marks)**

c Hence determine the coordinates of points where $y = 4$ intersects the curve. **(2 marks)**



Homework Exercise

13 Show that the line with equation $y = 2x - 5$ does not intersect the curve with parametric equations $x = 2t, \quad y = 4t(t - 1)$. **(4 marks)**

Problem-solving

Consider the discriminant after substituting.

14 The curve C has parametric equations $x = \sin t, \quad y = \cos 2t + 1, \quad 0 \leqslant t \leqslant 2\pi$.
Given that the line $y = k$, where k is a constant, intersects the curve,

a show that $0 \leqslant k \leqslant 2$ **(3 marks)**

b show that if the line $y = k$ is a tangent to the curve, then $k = 2$. **(3 marks)**

15 The curve C has parametric equations $x = e^{2t}, \quad y = e^{t} - 1$. The straight line l passes through the points A and B where $t = \ln 2$ and $t = \ln 3$ respectively.

a Find the points A and B. **(3 marks)**

b Show that the gradient of the line l is $\frac{1}{5}$ **(2 marks)**

c Hence, find the equation for line l in the form $ax + by + c = 0$. **(2 marks)**

16 The curve C has parametric equations $x = \sin t, \quad y = \cos t$. The straight line l passes through the points A and B where $t = \frac{\pi}{6}$ and $t = \frac{\pi}{2}$ respectively. Find an equation for the line l in the form $ax + by + c = 0$. **(7 marks)**

Homework Exercise

17 The diagram shows the curve C with parametric equations

$$x = \frac{t-1}{t}, \quad y = t - 4, \quad t \neq 0$$

The curve crosses the y-axis and the x-axis at points A and B respectively.

a Find the coordinates of A and B. **(4 marks)**

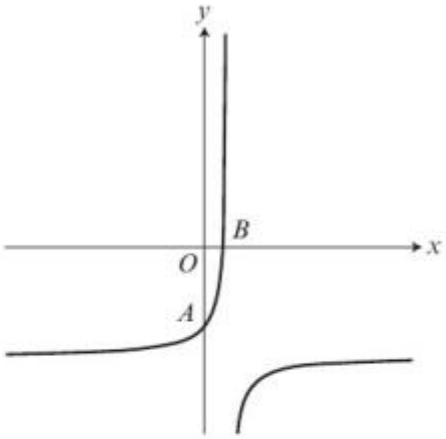
The line l_1 intersects the curve at points A and B. The lines l_2 and l_3 are parallel to l_1 and are distinct tangents to the curve.

b Show that the two possible equations for l_2 and l_3 are

$$y = 4x - 4 \text{ and } y = 4x - 12$$

(6 marks)

c Find the coordinates of the point where each tangent meets C. **(4 marks)**



Challenge

The curve C_1 has parametric equations

$$x = e^{2t}, \quad y = 2t + 1$$

The curve C_2 has parametric equations

$$x = e^{t}, \quad y = 1 + t^2$$

Find the coordinates of the points at which these two curves intersect.

Homework Answers

1 **a** $(11, 0)$ **b** $(7, 0)$ **c** $(1, 0), (9, 0)$
d $(1, 0), (2, 0)$ **e** $\left(\frac{9}{5}, 0\right)$

2 **a** $(0, -5)$ **b** $\left(0, \frac{9}{16}\right)$ **c** $(0, 0), (0, 12)$
d $\left(0, \frac{1}{2}\right)$ **e** $(0, 1)$

3 4 **4** 4 **5** $\left(\frac{1}{2}, \frac{3}{2}\right)$

6 $t = \frac{5}{2}$, $t = -\frac{3}{2}$; $\left(\frac{25}{4}, 5\right)$, $\left(\frac{9}{4}, -3\right)$

7 $(1, 2), (1, -2), (4, 4), (4, -4)$

8 **a** $\left(\frac{\pi^2}{4} - 1, 0\right), (0, \cos 1)$

b $\left(\frac{\sqrt{3}}{2}, 0\right), (0, 1)$

c $(1, 0)$

9 **a** $(e + 5, 0)$ **b** $(\ln 8, 0), (0, -63)$ **c** $\left(\frac{5}{4}, 0\right)$

10 $t = \frac{2}{3}$, $t = -1$, $\left(\frac{4}{9}, \frac{2}{3}\right)$, $(1, -1)$

11 $t = \frac{14}{5}$, $\left(\ln\frac{9}{5}, \ln\frac{3}{5}\right)$

12 **a** $\left(6\cos\left(\frac{\pi}{12}\right), 0\right), \left(6\cos\left(\frac{5\pi}{12}\right), 0\right)$

b $4\sin 2t + 2 = 4 \Rightarrow 4\sin 2t = 2 \Rightarrow \sin 2t = 0.5$

$2t = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow t = \frac{\pi}{12}, \frac{5\pi}{12}$

c $\left(6\cos\left(\frac{\pi}{12}\right), 4\right), \left(6\cos\left(\frac{5\pi}{12}\right), 4\right)$

13 $y = 2x - 5 \Rightarrow 4t(t - 1) = 2(2t) - 5 \Rightarrow 4t^2 - 8t + 5 = 0$
Discriminant $= 8^2 - 4 \times 4 \times 5 = 64 - 80 = -16 < 0$
Since the discriminant is less than 0, the quadratic has no solutions, therefore the two equations do not intersect.

14 **a** $\cos 2t + 1 = k$
max of $\cos 2t = 1$, so $k = 1 + 1 = 2$
min of $\cos 2t = -1$, so $k = -1 + 1 = 0$
Therefore, $0 \leqslant k \leqslant 2$

b $y = 1 - 2\sin^2 t + 1 = 2 - 2\sin^2 t = 2 - 2x^2$
$k = 2 - 2x^2 \Rightarrow 2x^2 + k - 2 = 0$
Tangent when discriminant $= 0$
Discriminant $= 0^2 - 4 \times 2 \times (k - 2) = 0$
$-8(k - 2) = 0 \Rightarrow k - 2 = 0 \Rightarrow k = 2$
Therefore, $y = k$ is a tangent to the curve when $k = 2$.

15 **a** $A(4, 1), B(9, 2)$ **b** Gradient of $l = \frac{2 - 1}{9 - 4} = \frac{1}{5}$

c $x - 5y + 1 = 0$

16 $y + \sqrt{3}x - \sqrt{3} = 0$

Homework Answers

17 **a** A(0, −3), B$\left(\frac{3}{4}, 0\right)$

b Gradient of $l_1 = 4$

Equation of l_2 and l_3: $y = 4x + c$

$t - 4 = \dfrac{4(t-1)}{t} + c \Rightarrow t^2 - 4t = 4t - 4 + ct$

$\Rightarrow t^2 - (8 + c)t + 4 = 0$

Tangent when discriminant = 0

$(-(8+c))^2 - 4 \times 1 \times 4 = 0$

$64 + 16c + c^2 - 16 = 0$

$c^2 + 16c + 48 = 0$

$(c + 4)(c + 12) = 0 \Rightarrow c = -4$ or $c = -12$

So, two possible equations for l_2 and l_3 are

$y = 4x - 4$ and $y = 4x - 12$

c $\left(\frac{1}{2}, -2\right)$, $\left(\frac{3}{2}, -6\right)$

Challenge

(1, 1), (e, 2)